Subject: che301 (Past Papers)

CR: Hafiza Mubeen

VU Medical Zone (biotechnologists)

Admins: HAfiza Mubeen, Muhammad Nouman, Tasha kHan, Iqra Shaheen

Write two example of Scatter diagram 2.

Scatter Diagram- To examine relationship between the two, paired, interrelated data types, To find whether or not 182 these two data types are interrelated. To determine how closely they are related to identify a problem point. Examples of Scatter diagram- Relationship between thermal treatment temperature of a steel material and its tensile strengths, Relationship between visit made by a salesman and volume of sales, Relationship between the number of persons visiting a department store and volume of sales, Relationship between absorbance and concentration of a coloured sample (UV spectroscopy); calibration curve, Relationship of volume to pressure or

temperature Scatter Diagram- Can identify cause and effect relation. Can understand the relationship between two results. END

What is Precipitation mechanism?

ANS: Precipitation occurs when the air becomes saturated with water vapor, and can no longer hold all of it in vapor form. This occurs when moist air cools, usually when an airmass rises through the atmosphere. However, an airmass can also cool (e.g. through radiative cooling, or ground contact with cold terrain) without a change in altitude.

Convective precipitation occurs when air rises vertically through the (temporarily) self-sustaining mechanism of convection. *Stratiform* precipitation occurs when large air masses rise diagonally as larger-scale atmospheric dynamics force them to move over each other. *Orographic* precipitation is similar, except the upwards motion is forced when a moving airmass encounters the rising slope of a landform such as a mountain ridge.

3). Non polar sorbant?

Ans: **Non polar sorbent (3)**

Non-Polar Sorbent:
• It is also called reverse phased.
• Where polar silica's ends are replaced with non-polar carbon chain.
• Same as in HPLC. Non-POLAR INTERACTIONS
• Also called hydrophobic or reverse phase
• Involves van der Waals / dispersion forces
• Sorbents - C2, C3, C4, iC4, tC4, C5, C6, C7, C8, C10, C12, C18, C20, C30 phenyl and cyclohexyl
• Applications - drugs of abuse, pesticides
• Analytes - protonated / neutral state, aromatics & alkyl chains
• Matrix - biologicals, water, aqueous buffers
• Elution solvents - typically non-polar to moderately polar

4. Affinity chromatography?

Ans: chromatography in which a macromolecule (such as a protein) is isolated and purified by passing it in solution through a column treated with a substance having a ligand for which the macromolecule has an affinity that causes it to be retained on the column.

5. Properties of precipitate?

Ans;

Properties of good precipitates
• Should be filtered and washed easily
• Very low solubility (no significant loss during filtration / washing)
• Uncreative with constituents of the atmosphere
• Chemical composition must be known, if necessarily ignited.

6. Explain auxochrome ?

ANS: The group of atoms which has capacity to alter the color of a chromophore is called as auxochrome. For example azobenzene is one of the most common examples of molecule with chromophore. These dyes contain two groups; the chromophore and the auxochrome.

All organic compounds are mainly composed of carbon and hydrogen atoms. They have certain functional groups which are responsible for the unique chemical and physical properties of organic compounds. For example, -NH2, -OH, -COOH, -CHO, >C=O are some examples of functional groups. The structure of organic compounds can be determine with the help of different techniques like X-ray diffraction, spectroscopy (NMR, IR), UV-spectrophotometry etc. Spectroscopy involves the interaction of organic compound with electromagnetic radiation.

The functional groups in the molecule interact with electromagnetic radiations. Molecules absorb certain wavelength of radiations and remaining amount can be detected in digital display to get complete information about the structure of molecule. The visible part of electromagnetic spectrum lies from 400 – 700 nm of wavelength. If any organic molecule absorbs light from visible region, it will be colorful and color of molecule depends on absorption of wavelength of radiations.

7. Equation of Beer's Lambert law. ?

Ans: **Beer Lambert Law**

Beer's Law may be written simply as:

$A = \varepsilon bc$

where A is absorbance (no units)
ε is the molar absorbtivity with units of L mol^{-1} cm^{-1} (formerly called the extinction coefficient)
b is the path length of the sample, usually expressed in cm
c is the concentration of the compound in solution, expressed in mol L^{-1}

8) Median with examples.?

To calculate Median arrange the replicates or your data in ascending or descending order.

1,3,3,6,7,8,9

MEDIAN IS=6

1,2,3,4,5,6,8,9

MEDIAN=(4+5)+2

=4.5

The middle of the reading in odd number of data is called median. If you have even number of data points then add the two mid readings and divideby two. Means take mean of the two middle reading. Mode means the most repeatable data point

9**). UV spectroscopy, principal and its applications.?**

ANS: **what is UV spectroscopy its principle and application?10**

Ultraviolet–visible **spectroscopy** or ultraviolet-visible**spectrophotometry** (**UV**-Vis or **UV**/Vis) refers to absorption **spectroscopy** or reflectance**spectroscopy** in the ultraviolet-visible spectral region. ... In this region of the electromagnetic spectrum, atoms and molecules undergo electronic transitions

Range:

Ultraviolet (**UV**) is an electromagnetic radiation with a wavelength from 10 nm to 400 nm, shorter than that of visible light but longer than X-rays.

Principle:

Molecules containing π-electrons or non-bonding electrons (n-electrons) can absorb the energy in the form of ultraviolet or visible light to excite these electrons to higher anti-bonding molecular orbitals.[2] The more easily excited the electrons (i.e. lower energy gap between the HOMO and the LUMO), the longer the wavelength of light it can absorb. There are four possible types of transitions (π-π*, n-π*, σ-σ*, and n-σ*), and they can be ordered as follows: σ-σ* > n-σ* > π-π* > n-π

Applications:

- Solutions of transition metal ions can be colored (i.e., absorb visible light) because d electrons within the metal atoms can be excited from one electronic state to another. The colour of metal ion solutions is strongly affected by the presence of other species, such as

certain anions or ligands. For instance, the colour of a dilute solution of copper sulfate is a very light blue; adding ammonia intensifies the colour and changes the wavelength of maximum absorption (λ_{max}).

- Organic compounds, especially those with a high degree of conjugation, also absorb light in the UV or visible regions of the electromagnetic spectrum. The solvents for these determinations are often water for water-soluble compounds, or ethanolfor organic-soluble compounds. (Organic solvents may have significant UV absorption; not all solvents are suitable for use in UV spectroscopy. Ethanol absorbs very weakly at most wavelengths.) Solvent polarity and pH can affect the absorption spectrum of an organic compound. Tyrosine, for example, increases in absorption maxima and molar extinction coefficient when pH increases from 6 to 13 or when solvent polarity decreases.
- While charge transfer complexes also give rise to colours, the colours are often too intense to be used for quantitative measurement.
- 10)**Tpyes of chromatography?**
  - ANS:: **Types of Chromatography**
    1. Adsorption
    2. Partition
    3. Ion-Exchange
    4. Molecular Exclusion
    5. Affinity

vu medical

- 11)**. Different type of analyte?**
- e. An analyte, component (in clinical chemistry) or chemical species, is a substance or chemical constituent that is of interest in an analytical procedure. Whilein chemical analysis, matrix refers to the components of a sample other than the analyte of interest. The matrix can have a considerable effect on the way the analysis is conducted and the quality of the results obtained; such effects are called matrixeffects?
- 12)**Chiral chromatography?**
- ANS: **Chiral column chromatography** is a variant of column chromatography in which the stationary phase contains a single enantiomer of a chiral compound rather than being achiral. The two enantiomers of the same analyte compound differ in affinity to the single-enantiomer stationary phase and therefore they exit the column at different times.
-
- **13) Different types of error. ?**

Ans: Errors can be divided into three categories:
(1) Personal Error
(2) Systematic Error
(3) Random Error

- **14)What is gravimetry and types.?**

ANS:Types of Gravimetric Analysis

There are 4 fundamental types of gravimetric analysis.

Volatilization gravimetry

Precipitation gravimetry

Electrogravimetry

Thermogravimetric

- **15)Classification of gevamatric methods? 5**
- Gravimetric methods are quantitative methods that are based on determining the mass of a pure compound to which the analyte is chemically related.
- **Gravimetric analysis**
  - Quantitative estimation
  - Traditional Method.
  - Cheap, easily available apparatus,
  - Slow and accurate
  - Wide range of sample conc.(ng- kg)
  - No calibration required
- **Classification of gravimetric analysis**
  1. Precipitation gravimetry
  2. Volatilization gravimetry
  3. Electrogravimetry

16)What do u know about the term Coplymeric sonbert?

Copolymeric sorbent:
• Hydrophobic & ionic retention mechanisms
• Reverse phase sorbent with cation or anion exchange
• Acidic, basic & neutral analyte applications
• Matrix - aqueous
• Elution solvents mixture of organics with acid or base

17).What energy change occur when an ion or molecule abosrbes light?

Absorption of light by molecules or ions causes three types of energy changes:
▪ **Rotational (Low energy change)**: molecule rotates around a center of gravity
▪ **Vibrational (Low energy change):** average internuclear distance of two or more atoms in the molecule changes
▪ **Electronic (High energy change)**: change in energy of the electrons or transitions

18)What is spectrophotometer?

19).What is UV spectrometry.. Write down its principle and applications 10 marks

20)Two types ko atomic emmision spectrometry ?

Types:

- 1Flame emission spectroscopy
- 2Inductively coupled plasma atomic emission spectroscopy
- 3Spark and arc atomic emission spectroscopy

21)Method Error?

• If a method say heat it stay at room temperature use buffer of pH 10
The meaning of such statements may be
different for different analyst

22)What is sample prepration ?

Ans: Sample preparations
• Gross samples may be huge
• Crushing, grinding, sieving, mixing, and dividing the sample (often into halves) to reduce its mass
• Digestion and Extraction techniques

23)What is MBC in detail? 3 marks

HPLC Columns

Columns, made of stainless steel of varying length and internal diameter (Ø) are used for analytical chromatography

- **Micro Bore Columns (MBC)**: Columns with internal diameters of less than 2 mm are often referred to as MBC. Mobile phase consumption 1/20 compared to that in 4.6 mm diameter. Small injection volume required.

24): which test can be applied for the analysis of variance?2

- Analysis of Variance
  • Z-test

• T-test
• F-Test

- **25): what kind of organic compaunds can change thier solubility?2**
- Organic acids include carboxylic acids (strong organic acids) and phenols (weak organic acids)
- Organic bases includes amines

26) what is fish bone?3

The second QC tool is cause and effect diagram… Solving a problem requires clarification of a cause and effect relationship, where the effect (e.g., the result of work) varies according to factors (e.g., facilities, machines used, method of work, workers, materials and parts used etc.).
• To obtain a good work result, we must identify the effects of various factors and develop measures to improve the result accordingly. Can obtain a clear overall picture of causal relation. (A change in the cause triggers a variation in the result.)

- Can list up all causes to identify important causes.
• Can determine the direction of action (counter measure).

Cause & Effect Diagram
• Also called fishbone or Ishikawa diagram presents a systematic representation of the relationship between the effect (result) and affecting factors (causes).

27: what is the fuction of alumina?3

Alumina (Al_2O_3) used as solid support, available in acidic basic and neutral phases

Alumina,
Polyamide (in TLC)

28)State uses of NMR spectroscopy and mass spectrometri?5

- **Uses of NMR spectroscopy.** Nuclear Magnetic Resonance (NMR) spectroscopy is an**analytical** chemistry technique used in quality control and reserach for determining the content and purity of a sample as well as its molecular structure.

- Nuclear magnetic resonance spectroscopy, most commonly known as **NMR**spectroscopy or magnetic resonance spectroscopy (MRS), is a research technique that exploits the magnetic properties of certain atomic nuclei. ... Thus, in organic chemistry practice, **NMR** analysis is used to confirm the identity of a substance.
-
- **Mass spectrometry:**
- **Mass spectrometry** (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their**mass**-to-charge ratio. In simpler terms, a **mass** spectrum measures the masses within a sample. **Mass spectrometry** is used in many different fields and is applied to pure samples as well as complex mixtures

29)give some applications of extractions scheme?3

Extraction Scheme
• We can Separate a complex mixture by designing an extraction scheme

- **Applications:**
  - For extraction of caffeine from tea leaves.
  - For extraction of nicotine from tobacco.

30)What is spectrophotometer?

an apparatus for measuring the intensity of light in a part of the spectrum, especially as transmitted or emitted by particular substances.

Spectrophotometer techniques are used to measure the concentration of solutes in solution by measuring the amount of the light that is absorbed by the solution in a cuvette placed in the **spectrophotometer** . 2. The **spectrophotometer** light intensity as a♦technique is to measures **function** of wavelength.

31)How Analytical chemistry is used for the environmental analysis (2)

In environmental analysis the polutents like effluent or waste water, effluent gases or stack emmissions, soil and water resources may be monitored to apply controls.
In clinical side body fluids including blood, urins, mucus and body tissues may be analysed for measurement of microbial load, drug or other attributes and certain enzymatic levels may be assessed

32)Source of error (2)

Sources of error
• Instrumental error

• Method error
• Human error

- **Instrumental Error**
  • If a balance weigh 2 g as 2.1 g.
  • If a 10 mL pipette draw a volume of 9.5 mL on average
- method error. If your analytical method have some limitations then it may generate error

Third type of errors are personnel error. Analyst error.

33)Chracteriatics Absorption band (5)

The two main characteristics of an absorption band are;
• **Its position:** Corresponds to wavelength of radiation whose energy is equal to that required for the electronic transition.
• Its **intensities**: Corresponds to the number of molecules which absorb the radiation of that wavelength, depends largely on probability of transition. It may be measured as Transmittance (T) or Absorbance (A). Note: If a compound contains more then one chromophore usually corresponds to the sum of the expected spectra for all the chromophores. If the chromophore are separated by at least two single bonds they behave independently. If two chromophores in the same molecule are identical, a single absorption band is observed

34)What is sample preparation (5)

Sample preparations
• Gross samples may be huge
• Crushing, grinding, sieving, mixing, and dividing the sample (often into halves) to reduce its mass
• Digestion and Extraction techniques